

Graphing $f(x) = |x|$ and Key Attributes

TEACHER KEY | Algebra II | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from a table and identify its domain, range, intercepts, symmetry, and minimum value.

Vocabulary

Absolute value — the distance a number is from **0**, always **non-negative** (≥ 0).

Vertex — the point where the two straight-line pieces of $f(x) = |x|$ meet: **(0, 0)**.

Key attributes — domain, range, **intercepts**, symmetry, and minimum/maximum.

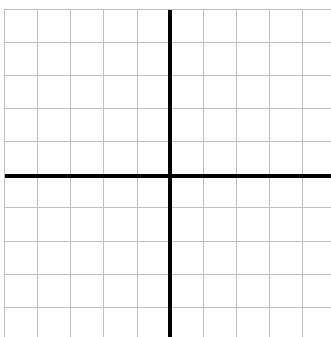
Symmetric about the y-axis — the left and right halves of the graph are **mirror images**.

Part 1 — Build the Table, Then Graph

Fill in $f(x) = |x|$ for each x -value.

x	-3	-2	-1	0	1	2	3
f(x)	3	2	1	0	1	2	3

Plot the seven points and connect them. Each square is 1 unit; both axes run -5 to 5.



Part 2 — Key Attributes of $f(x) = |x|$

Domain: **all real numbers, $(-\infty, \infty)$**

Range: **$[0, \infty)$**

x-intercept: **(0, 0)** y-intercept: **(0, 0)**

Vertex: **(0, 0)** Symmetric about: **the y-axis**

Minimum value: **0, at $x = 0$** Maximum value: **none**

Part 3 — What Does $|x|$ Mean in Real Life? Distance

A hiking trail runs both directions from a trailhead at position 0. A hiker's position is x miles (negative = one direction, positive = the other). The distance from the trailhead is $d(x) = |x|$.

Hiker at $x = -3.5$: distance = **3.5 miles**

Hiker at $x = 6$: distance = **6 miles**

A thermometer reads t degrees relative to 0°C (freezing). $|t|$ tells you how many degrees you are from freezing, above or below.

$t = -15$: **15** degrees from freezing $t = 8$: **8** degrees from freezing

In your own words, explain why distance can never come out negative.

Sample: **distance measures how far apart two things are, and "how far" is never a negative amount - that is exactly why the range of $|x|$ is $[0, \infty)$**

Part 4 — Compare to a Line

Fill in both columns to see what changes and what stays the same.

	$y = x$ (a line)	$f(x) = x $
Domain	all reals	all reals
Range	all reals	$[0, \infty)$
x- and y-intercept	$(0, 0)$	$(0, 0)$
Minimum?	none	0
Symmetric?	no	yes, about the y-axis

Practice

- $|-8| = 8$ 2. $|6| = 6$ 3. $|0| = 0$ 4. $|-2.5| = 2.5$
- If $f(x) = |x|$ and $x = -10$, $f(x) = 10$
- $f(x) = |x|$ has a vertex at $(0, 0)$. Give one more point on each side of it. Left: $(-1, 1)$ Right: $(1, 1)$
- True or False: the graph of $f(x) = |x|$ is symmetric about the x-axis. Explain.
False **it is symmetric about the y-axis, not the x-axis - x-axis symmetry would require negative outputs too, and $|x|$ never produces one**
- The points $(-4, 4)$ and $(4, 4)$ are both on the graph of $f(x) = |x|$. What key attribute does this illustrate?
Symmetry about the y-axis **points equally far from the vertex on each side share the same output**
- One point is at $x = -6$, a certain distance from zero. What x-value on the OTHER side of zero is the same distance away? **6**
- Which is larger, $|-15|$ or $|9|$? **$|-15|$, since $15 > 9$**
- Put these in order from least to greatest: $|-4|$, $|1|$, $|-9|$, $|6|$
 $|1|$, $|-4|$, $|6|$, $|-9|$ (that is: 1, 4, 6, 9)

Exit Ticket

For $f(x) = |x|$, give the domain, range, vertex, and axis of symmetry.

Domain **$(-\infty, \infty)$** Range **$[0, \infty)$**

Vertex **$(0, 0)$** Axis of symmetry **$x = 0$ (the y-axis)**